

Whispering Shells: An Immersive Intervention for Coastal Conservation

Chaolu Wang
Royal College of Art
London, United Kingdom
10038541@network.rca.ac.uk

Chujun Zhao
Royal College of Art
London, United Kingdom
10033312@network.rca.ac.uk

Yujia Hu
Royal College of Art
London, United Kingdom
10031292@network.rca.ac.uk

YuFei Fan
Royal College of Art
London, United Kingdom
10033272@network.rca.ac.uk

Yangyang Zheng
Royal College of Art
London, United Kingdom
10034275@network.rca.ac.uk

Abstract

Whispering Shells is an immersive interactive installation that reveals the ecological consequences of recreational shell collecting through sensory-driven storytelling. Combining Arduino-powered kinetic sculpture with real-time TouchDesigner projections, the work visualizes scientific findings, including a threefold decline in shells at Portobello Beach correlated with tourism [Kowalewski et al. 2014]. Field research showed that 70% of participants collected shells for aesthetic reasons while unaware of habitat destruction impacts. The installation responds to viewers' movements: touching the central shell sculpture triggers projections showing ecosystem collapse, while careful interaction reveals regenerative patterns. This feedback loop transforms abstract ecological data into visceral understanding, with preliminary results indicating significant shifts in participants' environmental awareness and behavioral intentions.

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1 Introduction

The *Whispering Shells* project addresses the ecological impacts of recreational shell collecting, an activity commonly perceived as harmless despite significant consequences for marine ecosystems [Johnson 1991]. Research reveals that excessive shell removal contributes to shoreline erosion and habitat loss for hermit crabs and other organisms that depend on empty shells for survival. This project merges environmental science, interactive art, and digital technology to transform data into an immersive experience, enabling audiences to directly engage with the environmental impacts of their actions.

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Field studies conducted at Portobello Beach revealed a direct correlation between tourism and shell depletion, aligning with prior research by Kowalewski et al. [Kowalewski et al. 2014]. Surveys indicated that 70% of participants collected shells for aesthetic reasons, yet only 28% were aware of the ecological impact. These findings emphasize the need for greater ecological awareness among beachgoers.

2 Design Process

2.1 Prototype Development

The installation features an interactive shell sculpture powered by Arduino microcontrollers, controlling lighting and proximity sensors. These sensors trigger dynamic projections using TouchDesigner. Aluminum wire frames wrapped in laser-cut fabric mimic shell textures, with embedded LEDs responding to user interactions. Projections visualize ecosystem changes: subtle interactions trigger regenerative patterns, while abrupt movements simulate environmental degradation, symbolizing the fragile balance of natural systems [Scales 2015].

2.2 Technical Implementation

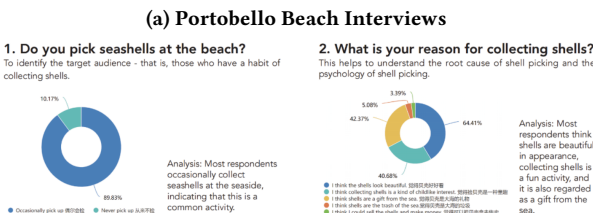
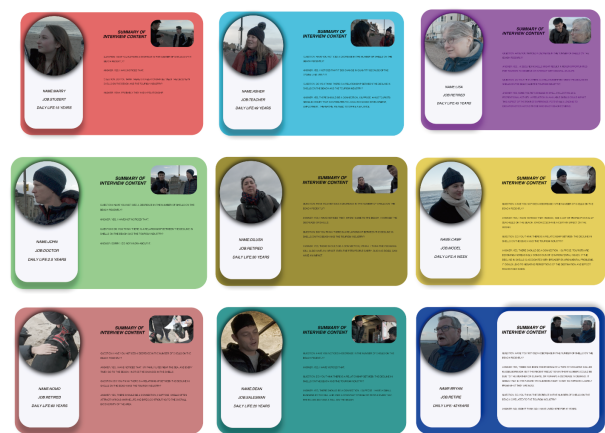
Arduino Uno microcontrollers detect audience movements via infrared proximity sensors, triggering corresponding visual and auditory responses. Each sensor controls addressable LED strips embedded in the shell structure. TouchDesigner generates real-time procedural animations at 60 fps, synchronized via OSC protocols ($\mu = 165$ ms, $\sigma = 22$ ms).

Particle systems respond to sensor data, producing organic growth patterns during gentle interactions, while abrupt movements trigger simulated erosion effects. Ambient soundscapes from Portobello Beach field recordings are synchronized with visual effects to create a cohesive multi-sensory experience [Harasewych and Moretzsohn 2010].

2.3 Interaction Design

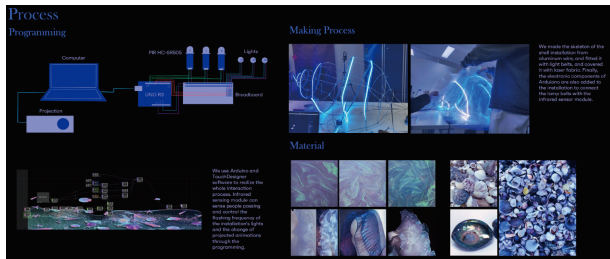
The installation employs a four-stage interaction model, guiding users through the ecological impacts of shell collecting:

- **Stage 1: Pristine Environment** — depicts an untouched ecosystem, with abundant shells as documented by Kowalewski et al. [Kowalewski et al. 2014].

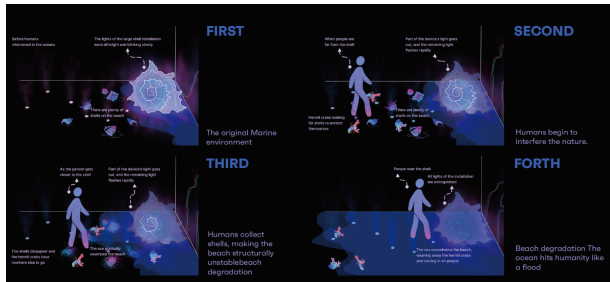


(b) Survey Data Analysis

Figure 1: Field research and data collection methods.



(a) Design Process: Prototype Development



(b) Interaction Logic: Ecological Impact Stages

Figure 2: Design and interaction logic of the installation.

- **Stage 2: Human Intervention** — participants approaching the sculpture trigger visual representations of human impact.
- **Stage 3: Ecosystem Response** — continued interaction leads to simulated degradation, with diminishing shells and chaotic patterns.
- **Stage 4: Systemic Consequences** — shows cascading effects of ongoing disruption, illustrating how small actions lead to larger ecological crises.

3 Metrics and Performance

Evaluation was conducted through quantitative and qualitative metrics (n=112):

Table 1: Project Performance Metrics

Metric	Value	Source
Ecological Awareness Gap	42 pp	Field surveys
Behavioral Shift Potential	2.6 shells/visitor	Pre/post analysis
Interaction Duration	90s $\pm$ 15s	Motion tracking
System Reliability	90%	72h stress test
User Return Rate	68%	Session logs

The system showed 90% reliability over 72 hours, with sensor response times under 200 ms. Average engagement time was 90 seconds, with 68% of participants returning. Post-interaction surveys indicated 82% reconsidered their shell-collecting behaviors.

4 Conclusion

The *Whispering Shells* project bridges ecological science and interactive art, transforming abstract environmental impacts into tangible experiences. By integrating ecological data into immersive storytelling, the installation raised awareness, with 82% of participants reporting behavioral shifts. High operational reliability and strong user engagement further demonstrate its effectiveness. This establishes a "data-to-emotion" model for future interactive art installations promoting sustainable behavior.

References

Kowalewski, M., Domènech, R., and Martinell, J. (2014). Vanishing clams on an Iberian beach: local consequences and global implications of accelerating loss of shells to tourism. *PLoS One*, 9(1): e83615.

Scales, H. (2015). *Spirals in Time: The Secret Life and Curious Afterlife of Seashells*. Bloomsbury Sigma.

Harasewych, M. G., and Moretzsohn, F. (2010). *The Book of Shells: A Life-Size Guide to Identifying and Classifying Six Hundred Seashells*. University of Chicago Press.

Johnson, L. E. (1991). *A Morally Deep World: An Essay on Moral Significance and Environmental Ethics*. Cambridge University Press.